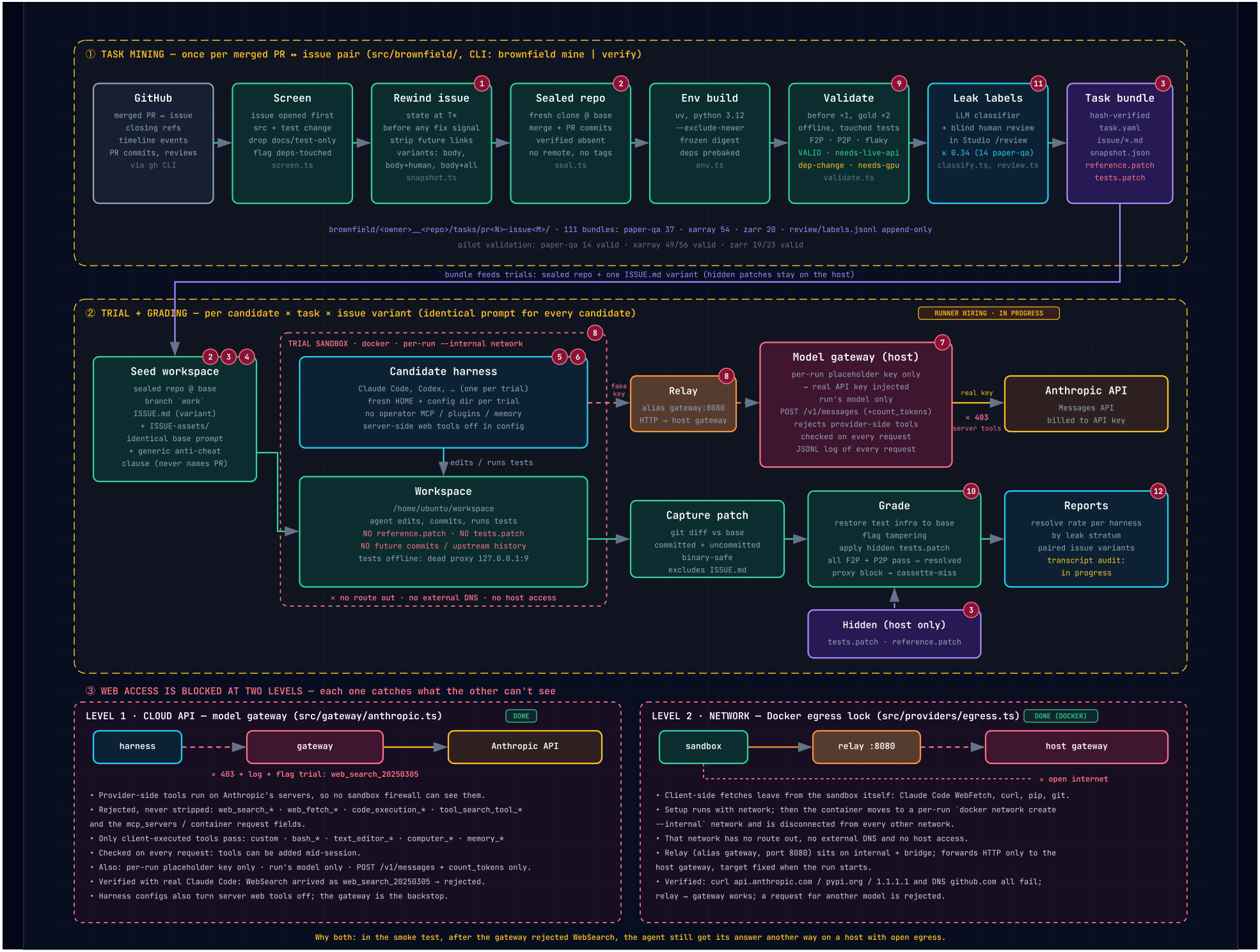
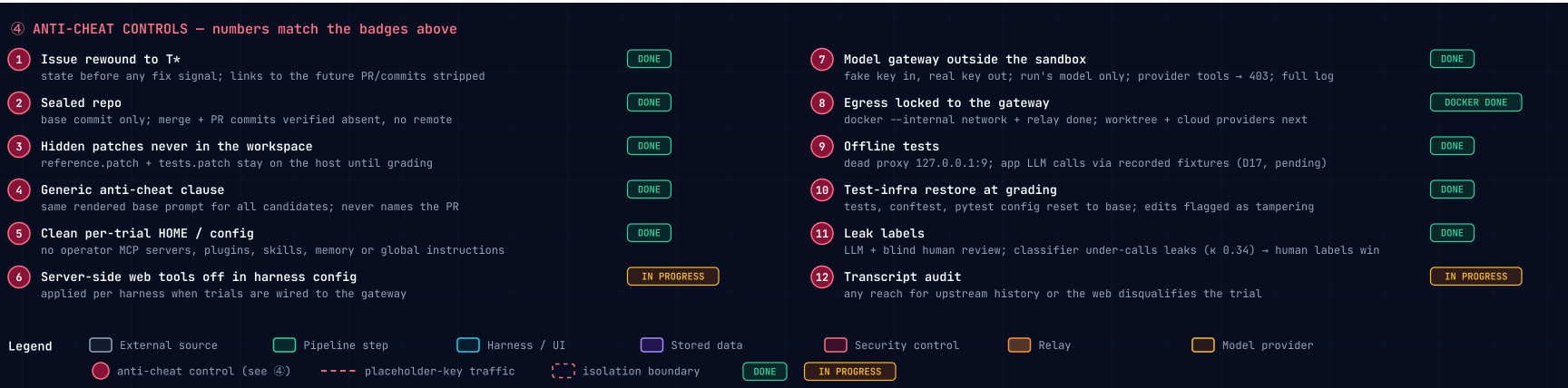

① TASK MINING — once per merged PR ↔ issue pair (src/brownfield/, CLI: brownfield mine | verify)
GitHub
merged PR ↔ issue
closing refs
timeline events
PR commits, reviews
via gh CLI
Screen
issue opened first
src + test change
drop docs/test-only
flag deps-touched
screen.ts
Rewind issue
1
state at T*
before any fix signal
strip future links
variants: body,
body+human, body+all
snapshot.ts
Sealed repo
2
fresh clone @ base
merge + PR commits
verified absent
no remote, no tags
seal.ts
Env build
uv, python 3.12
--exclude-newer
frozen digest
deps prebaked
env.ts
Validate
9
before ×1, gold ×2
offline, touched tests
F2P · P2P · flaky
VALID · needs-live-api
dep-change · needs-gpu
validate.ts
Leak labels
11
LLM classifier
+ blind human review
in Studio /review
κ 0.34 (14 paper-qa)
classify.ts, review.ts
Task bundle
3
hash-verified
task.yaml
issue/*.md
snapshot.json
reference.patch
tests.patch
brownfield/<owner>__<repo>/tasks/pr<N>-issue<M>/ · 111 bundles: paper-qa 37 · xarray 54 · zarr 20 · review/labels.jsonl append-only
pilot validation: paper-qa 14 valid · xarray 49/56 valid · zarr 19/23 valid
bundle feeds trials: sealed repo + one ISSUE.md variant (hidden patches stay on the host)
② TRIAL + GRADING — per candidate × task × issue variant (identical prompt for every candidate)
RUNNER WIRING · IN PROGRESS
Seed workspace
2 3 4
sealed repo @ base
branch `work`
ISSUE.md (variant)
+ ISSUE-assets/
identical base prompt
+ generic anti-cheat
clause (never names PR)
TRIAL SANDBOX · docker · per-run --internal network
8
Candidate harness
5 6
Claude Code, Codex, … (one per trial)
fresh HOME + config dir per trial
no operator MCP / plugins / memory
server-side web tools off in config
edits / runs tests
Workspace
/home/ubuntu/workspace
agent edits, commits, runs tests
NO reference.patch · NO tests.patch
NO future commits / upstream history
tests offline: dead proxy 127.0.0.1:9
× no route out · no external DNS · no host access
fake key
Relay
8
alias gateway:8080
HTTP → host gateway
Model gateway (host)
7
per-run placeholder key only
→ real API key injected
run's model only
POST /v1/messages (+count_tokens)
rejects provider-side tools
checked on every request
JSONL log of every request
real key
× 403
server tools
Anthropic API
Messages API
billed to API key
Capture patch
git diff vs base
committed + uncommitted
binary-safe
excludes ISSUE.md
Grade
10
restore test infra to base
flag tampering
apply hidden tests.patch
all F2P + P2P pass → resolved
proxy block → cassette-miss
Reports
12
resolve rate per harness
by leak stratum
paired issue variants
transcript audit:
in progress
Hidden (host only)
3
tests.patch · reference.patch
③ WEB ACCESS IS BLOCKED AT TWO LEVELS — each one catches what the other can't see
LEVEL 1 · CLOUD API — model gateway (src/gateway/anthropic.ts)
DONE
harness
gateway
Anthropic API
× 403 + log + flag trial: web_search_20250305
• Provider-side tools run on Anthropic's servers, so no sandbox firewall can see them.
• Rejected, never stripped: web_search_* · web_fetch_* · code_execution_* · tool_search_tool_*
and the mcp_servers / container request fields.
• Only client-executed tools pass: custom · bash_* · text_editor_* · computer_* · memory_*
• Checked on every request: tools can be added mid-session.
• Also: per-run placeholder key only · run's model only · POST /v1/messages + count_tokens only.
• Verified with real Claude Code: WebSearch arrived as web_search_20250305 → rejected.
• Harness configs also turn server web tools off; the gateway is the backstop.
LEVEL 2 · NETWORK — Docker egress lock (src/providers/egress.ts)
DONE (DOCKER)
sandbox
relay :8080
host gateway
× open internet
• Client-side fetches leave from the sandbox itself: Claude Code WebFetch, curl, pip, git.
• Setup runs with network; then the container moves to a per-run `docker network create
--internal` network and is disconnected from every other network.
• That network has no route out, no external DNS and no host access.
• Relay (alias gateway, port 8080) sits on internal + bridge; forwards HTTP only to the
host gateway, target fixed when the run starts.
• Verified: curl api.anthropic.com / pypi.org / 1.1.1.1 and DNS github.com all fail;
relay → gateway works; a request for another model is rejected.
Why both: in the smoke test, after the gateway rejected WebSearch, the agent still got its answer another way on a host with open egress.



How a task is made

- Take a merged PR that closes an issue opened before the fix work began
- Rewind the issue to T* and seal the repo at the PR's base commit
- Validate offline: the gold patch must turn hidden tests from failing to passing (F2P) without breaking others (P2P), twice
- Label how much each issue body/comment leaks the fix (LLM + blind human review)

How a harness is scored

- Every candidate gets the same sealed repo, ISSUE.md variant and base prompt
- Its diff vs base is captured; test files and pytest config are reset to base
- Hidden tests.patch applied; resolved only if every F2P and P2P test passes
- Results reported by leak stratum and paired across issue variants

Status (2026-11-01)

- Mining, validation (touched tests), bundles, Studio review: done
- Grading, workspace seeding, patch capture: done
- Gateway core + Docker egress lock: done and verified
- In progress: wiring into the trial runner, worktree/cloud egress, transcript audit
- Full-suite validation deferred to the cloud orchestrator